

181. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500

181.1 Overview

This clause specifies the 800GBASE-FR4-500 Physical Medium Dependent (PMD) type together with the single-mode fiber medium. The optical signals generated by this PMD type are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMD provides point-to-point 800 Gb/s Ethernet links over four wavelength division multiplexing (WDM) lanes on single-mode fiber, with a reach of up to at least 500 m.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 181–1 to the medium through the medium dependent interface (MDI) and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Table 181–1—Physical Layer clauses associated with the 800GBASE-FR4-500 PMDs

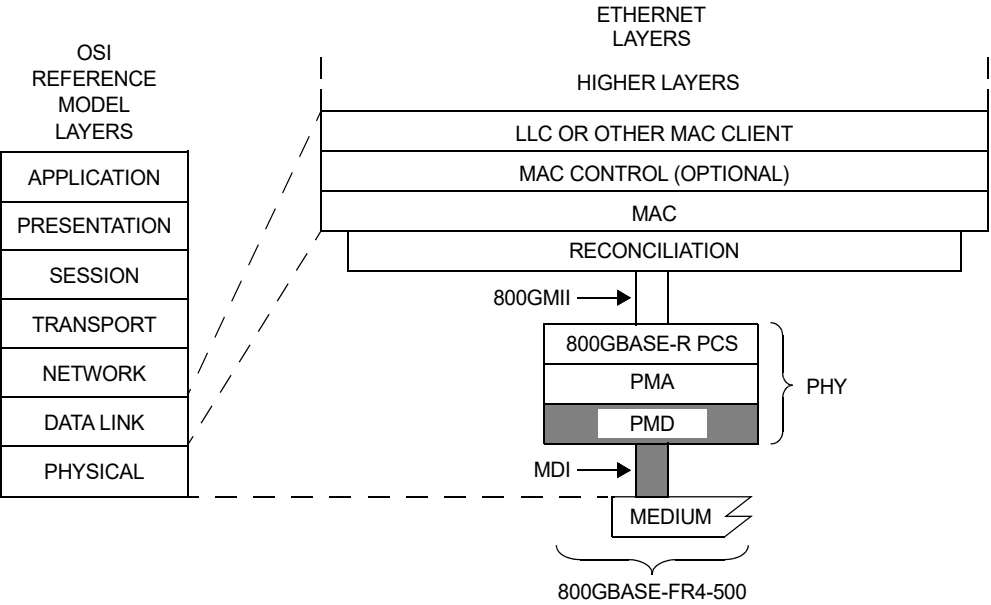
Associated clause	800GBASE-FR4-500
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a

^a One or two 800GAUI-n may be instantiated within a 800GBASE-FR4-500 PHY as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Figure 181–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
FR4-500 = PMD FOR SINGLE-MODE FIBER — 500 m

Figure 181–1—800GBASE-FR4-500 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

181.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA adjacent to the PMD using the method described in 174A.9, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 3.2×10^{-5} .

181.3 PMD service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (PMA immediately above the PMD). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 169.3. For PMD:IS_UNITDATA_i.request primitive in the transmit direction and PMD:IS_UNITDATA_i.indication primitive in the receive direction, where $i = 0$ to 3, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 106.25 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive (see 116.3.3.3) corresponds to the variable rts_status of the ready-to-send (RTS) function (see 181.5.12), as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication primitive on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the PMD:IS_SIGNAL.request primitive (see 116.3.3.4).

181.4 Delay and Skew

181.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-FR4-500 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

181.4.2 Skew constraints

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5. The measurements of Skew and Skew Variation are defined in 86.8.3.1.

181.5 PMD functional specifications

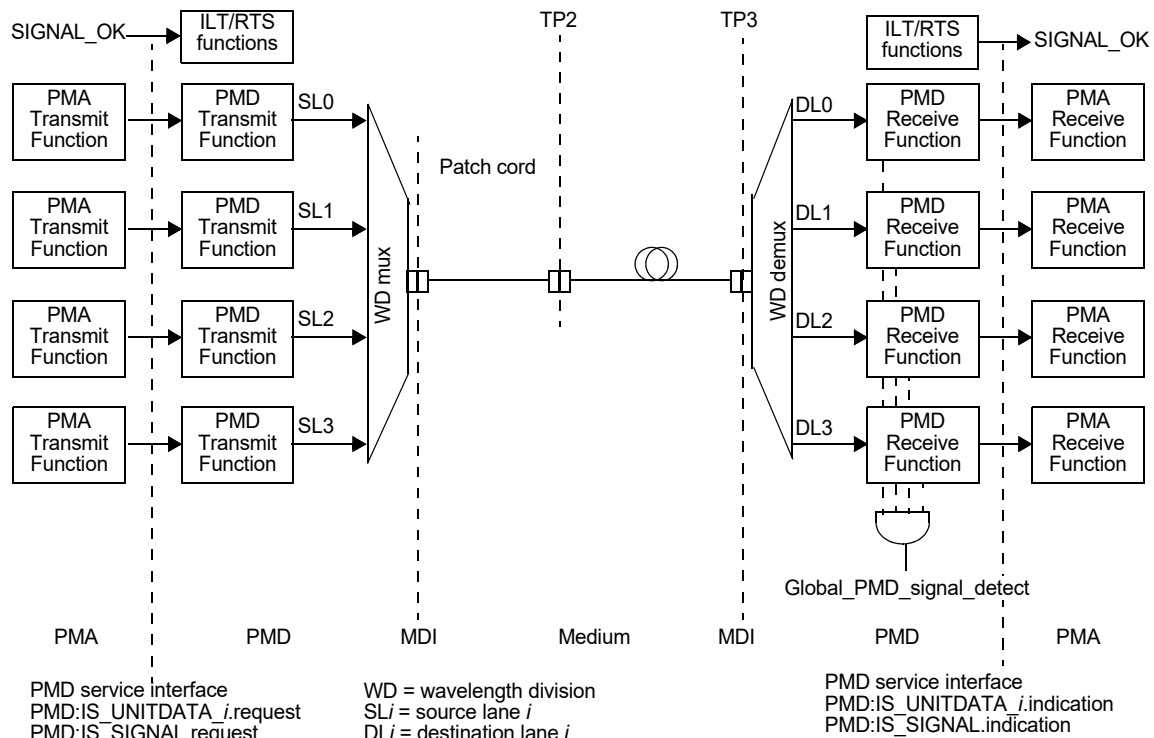
The 800GBASE-FR4-500 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

181.5.1 Block diagram

The block diagram is shown in Figure 181–2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 181.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 181.8.3). Unless specified otherwise, all receiver measurements and tests defined in 181.9 are made at TP3.

While TP2 and the patch cord used for transmitter measurements and tests are shown in Figure 181–2, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.

The inter-sublayer link training (ILT) and RTS functions indicated in Figure 181–2 are defined in Annex 178B.



NOTE: For clarity, only one direction of transmission is shown

Figure 181–2—Block diagram for 800GBASE-FR4-500 transmit/receive paths

181.5.2 PMD transmit function

The PMD transmit function has two operating modes: DATA and TRAINING. The operating mode is controlled by the tx_mode variable of the ILT function (see 181.5.12): it is DATA when tx_mode = DATA, and TRAINING otherwise.

When operating in DATA mode, the PMD transmit function shall convert the four symbol streams requested by the PMD service interface PMD:IS_UNITDATA₀.request primitive to PMD:IS_UNITDATA₃.request primitive into four separate optical signals. The four optical signals shall then be wavelength division multiplexed and delivered to the MDI, all according to the transmit optical specifications in this clause.

The polarity of the PMD output on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI output, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI output, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 178B.7.7). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

When operating in TRAINING mode, the PAM4 symbol stream on each lane is provided by the ILT function (see 178B.7).

181.5.3 PMD receive function

The PMD Receive function shall demultiplex the composite optical signal received from the MDI into four separate optical signals. The four optical signals shall then be converted into four symbol streams for delivery to the PMD service interface using the PMD:IS_UNITDATA_0.indication primitive to PMD:IS_UNITDATA_3.indication primitive, all according to the receive optical specifications in this clause.

The polarity of the PMD input on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI input, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI input, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.7.7). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

181.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all 4 lanes and it is set to the logical-and of PMD_signal_detect_i from all of the PMD lanes.

181.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane *i*, where *i* is in the range 0 to 3, a receiver shall set the variable PMD_signal_detect_i to either 0 or 1 according to the criteria in Table 181–2

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_*i* variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 181–2—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_ <i>i</i>
Average optical power at TP3 ≤ -16 dBm	0
For all lanes; [The input signal is a compliant 800GBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 181–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

181.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the ILT and RTS functions associated with the PMD (see 181.5.12).

181.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the Global_PMD_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 181–5.

If a PMD_fault is detected, then the PMD may set the Global_PMD_transmit_disable variable to one, turning off the optical transmitter in each lane.

181.5.8 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function allows the optical transmitters in each lane to be selectively disabled. This function is used by the ILT function. It may optionally be provided through control variables PMD_transmit_disable_ *i*.

If either PMD_transmit_disable_ *i* or the variable tx_disable in lane *i* of the ILT function (see 178B.8.2.1) is set to one (where *i* represents the lane number in the range 0:*n*-1), this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 181–5.

If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_ *i* variable to one, turning off the optical transmitter in each lane.

181.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

181.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

181.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

181.5.12 Autonomous path startup (APSU) functions

A PMD shall provide the RTS function and the ILT function with a training frame of O1 format, specified in Annex 178B.

When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

When tx_mode = LOCAL_PATTERN (see 178B.8.3.1) the PMD transmits PRBS31Q (see 176.7.4.2).

The default value of max_wait_timer_duration is 60, corresponding to a duration of 60 seconds for max_wait_timer (see 178B.8.3.3).

181.6 Wavelength-division-multiplexed lane assignments

The wavelength range for each lane of the 800GBASE-FR4-500 is defined in Table 181–3. The 800GBASE-FR4-500 center wavelengths are members of the CWDM wavelength grid defined in ITU-T G.694.2 and are spaced at 20 nm.

Table 181–3—800GBASE-FR4-500 wavelength-division-multiplexed lane assignments

Lane	Center wavelength	Wavelength range
0	1271 nm	1264.5 to 1277.5 nm
1	1291 nm	1284.5 to 1297.5 nm
2	1311 nm	1304.5 to 1317.5 nm
3	1331 nm	1324.5 to 1337.5 nm

NOTE—Each functional lane, denoted SLi at the transmitter and DLi at the receiver, is mapped to a specific wavelength to support ILT operation.

181.7 PMD to MDI optical specifications

The operating range for the PMD is defined in Table 181–4. A 800GBASE-FR4-500 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-FR4-500 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Table 181–4—800GBASE-FR4-500 operating range

PMD type	Required operating range
800GBASE-FR4-500	2 m to 500 m

181.7.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 181–5 per the definitions in 181.9.

Table 181–5—800GBASE-FR4-500 transmit characteristics

Description	Reference	800GBASE-FR4-500	Unit
Signaling rate, each lane (range)	—	106.25 ± 50 ppm	GBd
Modulation format	—	PAM4	—
Lane wavelengths (range)	181.9.3	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	nm
Side-mode suppression ratio (SMSR), each lane (min)	181.9.3	30	dB
Total average launch power (max)	181.9.4	10.9	dBm
Average launch power, each lane (max)		4.9	dBm
Average launch power, each lane ^a (min)		–2.1 ^b	dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (max)	181.9.5	4.8	dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (min) for $\max(\text{TECQ}, \text{TDECQ}) < 0.9$ dB for $0.9 \text{ dB} \leq \max(\text{TECQ}, \text{TDECQ}) \leq 3.4$ dB		0.9 $\max(\text{TECQ}, \text{TDECQ})$	dBm dBm
Difference in launch power between any two lanes (OMA _{outer}) (max)	—	4	dB
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	181.9.6	3.4	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	181.9.7	3.4	dB
$ \text{TDECQ} - \text{TECQ} $, each lane (max)	—	2.5	dB
Transmitter functional symbol error histogram, each lane (max)	181.9.8	See Table 180–17	—
Transmitter overshoot and undershoot, each lane (max)	181.9.9	27	%
Transmitter power excursion, each lane (max)	181.9.10	3.5	dBm
Extinction ratio, each lane (min)	181.9.11	3.5	dB
Transmitter transition time, each lane (max)	181.9.12	8	ps
Average launch power of OFF transmitter, each lane (max)	—	–15	dBm
RIN _x OMA, each lane (max)	181.9.13	–139	dB/Hz
Optical return loss tolerance (max)	—	17.1	dB
Transmitter reflectance ^c (max)	—	–26	dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of –2.1 dBm corresponds to an OMA of 0.9 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 181–5 vary with $\max(\text{TECQ}, \text{TDECQ})$. The relationships are illustrated in Figure 181–3 along with the values for OMA_{outer} each lane (max).

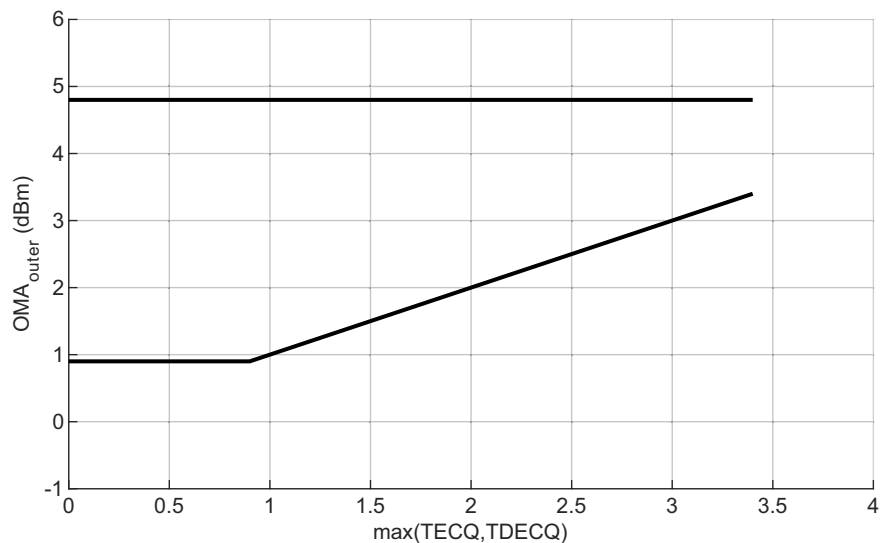


Figure 181-3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max(TECQ, TDECQ)

181.7.2 800GBASE-FR4-500 receive optical specifications

The receiver shall meet the specifications in Table 181-6 per the definitions in 181.9.

Table 181–6—800GBASE-FR4-500 receive characteristics

Description	Reference	800GBASE-FR4-500	Unit
Signaling rate, each lane (range)	—	106.25 ± 50 ppm	GBd
Modulation format	—	PAM4	—
Lane wavelengths (range)	181.9.3	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	nm
Damage threshold ^a , each lane	—	5.9	dBm
Average receive power, each lane (max)	—	4.9	dBm
Average receive power, each lane ^b (min)	—	–5.6	dBm
Receive power (OMA _{outer}), each lane (max)	—	4.8	dBm
Difference in receive power between any two lanes (OMA _{outer}) (max)	—	4.1	dB
Receiver reflectance (max)	—	–26	dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max) for TECQ < 0.9 dB for 0.9 dB ≤ TECQ ≤ SECQ	181.9.14	–3.2 –4.1 + TECQ	dBm dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	181.9.15	–0.7	dBm
Conditions of stressed receiver sensitivity test: ^d			
Stressed eye closure for PAM4 (SECQ), lane under test	—	3.4	dB
OMA _{outer} of each aggressor lane	—	1.3	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 181.8) for the BLER specified in 181.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 181–6 vary depending on the value of TECQ for the test source used for measurement. Figure 181–4 illustrates this.

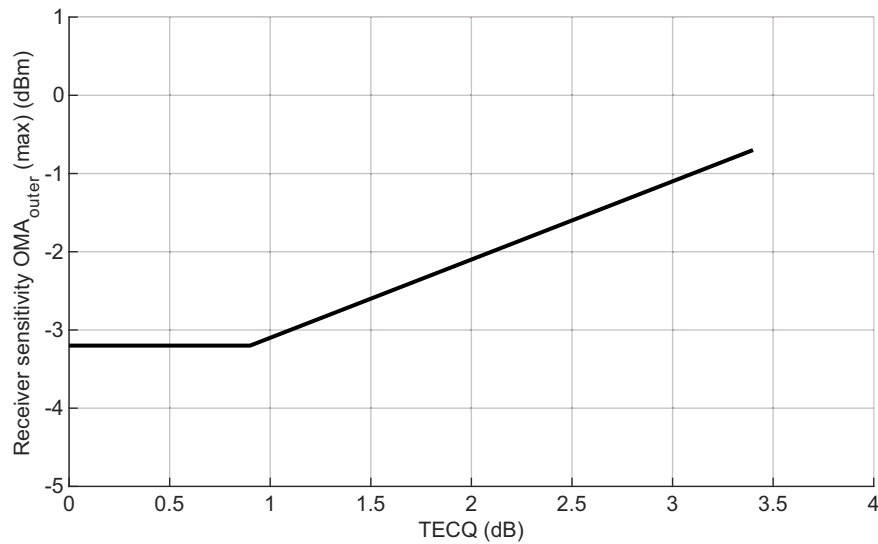


Figure 181-4—Receiver sensitivity (OMA_{outer}), each lane (max)

181.7.3 800GBASE-FR4-500 illustrative link power budget

Illustrative power budgets and penalties for the channels are shown in Table 181-7.

Table 181-7—800GBASE-FR4-500 illustrative link power budgets

Parameter	800GBASE-FR4-500	Unit
Power budget (for maximum TDECQ)	7.5	dB
Operating distance	500	m
Channel insertion loss ^{a, b}	3.5	dB
Maximum discrete reflectance	−35	dB
Allocation for penalties ^c (for maximum TDECQ)	4	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 181-4 and cabled optical fiber attenuation of 0.5 dB/km plus an allocation for connection and splice loss given in 181.8.2.

^b The channel insertion loss of 3.5 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.4 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.2 dB depending on the discrete reflectances in the link as shown in Table 181-10.

^c Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. In this illustrative link power budget, the value shown includes an allocation of 0.4 dB for MPI penalty and 0.2 dB for DGD penalty. For cases with a channel insertion loss less than 3.5 dB, as shown in Table 181-10, the allocation for penalties should be “7.5 – channel insertion loss”.

The values of transmitter $\text{OMA}_{\text{outer}}$ (max), transmitter $\text{OMA}_{\text{outer}}$ (min) versus $\max(\text{TECQ}, \text{TDECQ})$, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) (max) versus TECQ are illustrated in Figure 181–5.

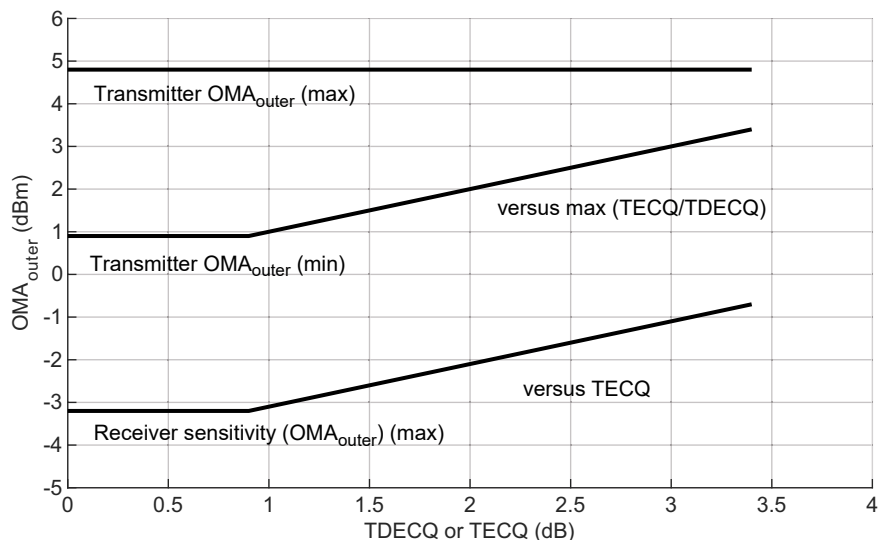


Figure 181–5—Transmitter $\text{OMA}_{\text{outer}}$ each lane versus $\max(\text{TECQ}, \text{TDECQ})$ and receiver sensitivity ($\text{OMA}_{\text{outer}}$) each lane versus TECQ

181.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 181–6.

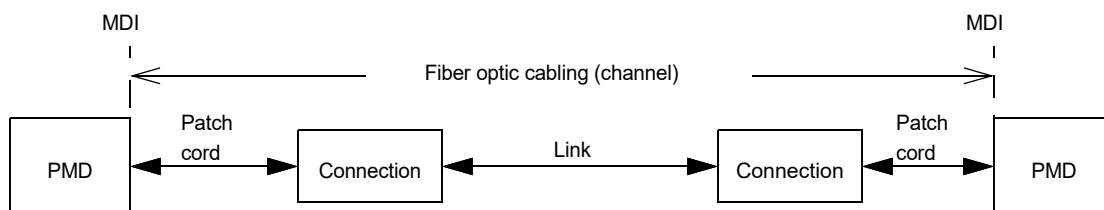


Figure 181–6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 181–8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. The channel insertion loss is given in Table 181–8. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 181–8—Optical channel characteristics

Description	800GBASE-FR4-500	Unit
Operating distance (max)	500	m
Channel insertion loss ^{a, b} (max)	See Table 181–10	dB
Channel insertion loss (min)	0	dB
Positive dispersion ^b (max)	1.5	ps/nm
Negative dispersion ^b (min)	–2.82	ps/nm
DGD_max ^c	2.24	ps
Optical return loss (min)	27	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1264.5 nm to 1337.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

181.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables meeting the characteristics in Table 181–9. The use of optical fiber cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), type G.657.A1, or type G.657.A2 (bend insensitive) fibers is recommended.

Table 181–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in TIA 568-C.3.

181.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 181–6, consists of a mated pair of optical connectors.

The maximum link distance is based on an allocation of 0.25 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 181–10 being allocated to connection and splice loss. For example, with no discrete reflections greater than –55 dB, a total connection and splice

loss allocation of 3.25 dB supports six connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 181–8 are met.

Table 181–10—Maximum channel insertion loss versus number of discrete reflectances

Maximum channel insertion loss (dB)		Number of discrete reflectances > −55 dB and ≤ −45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectances > −45 dB and ≤ −35 dB	0	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	1	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	2	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	3	3.5	3.5	3.5	3.5	3.5	3.5	3.4	3.4	3.4
	4	3.5	3.5	3.4	3.4	3.4	3.3	3.3	3.3	— ^a
	5	3.3	3.3	3.3	3.3	— ^a	— ^a	— ^a	— ^a	— ^a
	6	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

181.8.3 Medium Dependent Interface (MDI)

The 800GBASE-FR4-500 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 181–6). Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 181.5.1, not at the MDI.

181.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

181.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 181–12 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 181–12 may be used to perform that test. The test patterns used in this clause are shown in Table 181–11.

Table 181–11—Test patterns

Pattern	Pattern description	Defined in
3 ^a	PRBS31Q	176.7.4.2
4	PRBS13Q	176.7.4.3
5 ^a	Scrambled idle	172.2.4.11
6	SSPRQ	176.7.4.5
7	Valid 800GBASE-R signal	172.2.4

^a Precoding (see 176.7.1.2) is added to this pattern as necessary (see 181.9.14 and 181.9.15).

Table 181–12—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	3, 4, 5, 6 or 7	181.9.3
Side mode suppression ratio	3, 5, 6 or 7	181.9.3
Average optical power	3, 5, 6 or 7	181.9.4
Outer optical modulation amplitude (OMA _{outer})	4 or 6	181.9.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	181.9.6
Transmitter eye closure for PAM4 (TECQ)	6	181.9.7
Transmitter functional test	3	181.9.8
Transmitter overshoot and undershoot	6	181.9.9
Transmitter power excursion	6	181.9.10
Extinction ratio	4 or 6	181.9.11
Transmitter transition time	6	181.9.12
RIN _{17.1} OMA	4 or 6	181.9.13
Receiver sensitivity	3	181.9.14
Stressed receiver conformance test signal calibration	6	181.9.15
Stressed receiver sensitivity	3	181.9.15

181.9.2 Reference receiver

A reference receiver is defined for transmitter measurements. The reference receiver is specified in 180.9.2.

181.9.3 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 181–5 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns defined in Table 181–12.

181.9.4 Average optical power

The average optical power of each lane shall be within the limits given in Table 181–5 if measured using the methods given in IEC 61280-1-1, with the sum of the optical power from all of the lanes not under test below –30 dBm, per the test setup in Figure 53–6. The average optical power is measured using one of the test patterns defined in Table 181–12.

181.9.5 Outer optical modulation amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 181–5. OMA_{outer} is measured using one of the test patterns specified for OMA_{outer} in Table 181–12 as the difference between the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 181–7. OMA_{outer} is measured using the waveforms captured at the output of the reference receiver defined in 181.9.2.

When measured with pattern 6, OMA_{outer} is measured as the difference between the average optical launch power level P_3 , measured over the central 2 UI of the first 7 UI of all runs of 7 or more threes, and the average optical launch power level P_0 , measured over the central 2 UI of the first 6 UI of all runs of 6 or more zeros.

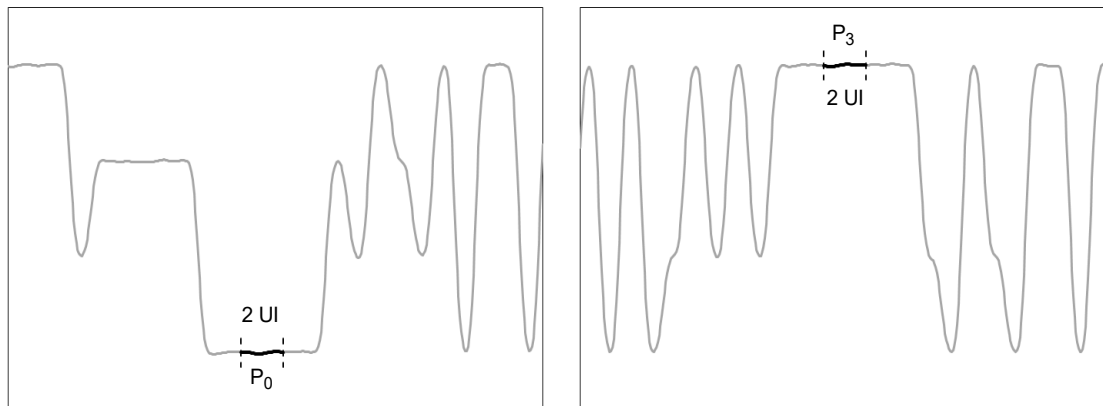


Figure 181–7—Example power levels P_0 and P_3 from PRBS13Q test pattern

181.9.6 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limits given in Table 181–5 if measured using the methods specified in 180.9.6 with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 181–5 and uses the test pattern specified for TDECQ in Table 181–12.
- For those cases where there is an 800GAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 181–1), the clock source for the SSPRQ test pattern is derived from the clock recovered

from the 800GAUI-n input signal. The pattern of the 800GAUI-n input signal may be PRBS31Q, scrambled idle, or a valid 800GBASE-R signal.

- The transmitter is tested using a worst case optical channel specified in 181.9.6.1.
- The conformance test setup uses the same block diagram as shown in Figure 180–9 with the exception that a demultiplexing filter is used in front of the reference receiver to separate the optical signal of the lane under test.

181.9.6.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 181–13.

Table 181–13—Transmitter compliance channel specifications

Lane	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
	Minimum	Maximum			
0	$0.0508 \times (\lambda - 1271) - 2.49$	$0.0468 \times (\lambda - 1271) - 1.45$	Minimum	17.1	0.5
1	$0.0485 \times (\lambda - 1291) - 1.5$	$0.0458 \times (\lambda - 1291) - 0.53$			
2	$0.0463 \times (\lambda - 1311) - 0.55$	$0.0443 \times (\lambda - 1311) + 0.37$			
3	$0.044 \times (\lambda - 1331) + 0.35$	$0.0423 \times (\lambda - 1331) + 1.23$			

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 500 m distance.

^b The value “Minimum” implies that the test channel insertion loss does not excessively degrade the signal to noise ratio of the measurement.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A 800GBASE-FR4-500 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 181–13 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 181–13. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 181–13.

181.9.7 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter eye closure at TP2. The TECQ of each lane shall be within the limit given in Table 181–5 if measured using the test pattern specified for TECQ in Table 181–12. The TECQ of each lane is measured using the methods specified for TDECQ in 181.9.6, except that the test fiber is not used.

181.9.8 Transmitter functional test

The transmitter functional symbol error histogram is measured using the method defined in 180.9.8 with the following exceptions:

- The transmitter under test is connected to the functional receiver (FnRx) by a test fiber, which meets the requirements in 181.9.6.1. Full compliance with 181.9.6.1 requires the test fiber to provide back reflection with polarization state resulting in the greatest RIN. Other test fibers, including patch cords, may be used for additional tests.

where in 180.9.8.1

- The FnRx is a variable optical attenuator (VOA) followed by an optical receiver (ORx) that complies with characteristics as given in Table 181–6. The VOA and test fiber approximate the compliance channel as given in Table 181–13 with the exception that the channel loss is not minimum.

where in Equation (180–16)

- $Tx_DUT_power_budget$ is the transmitter under test power budget for the specific test fiber as given in Table 181–7, except uses measured instead of max TDECQ value and is given by Equation (180–17).

where in Equation (180–17)

- $Channel_insertion_loss$ is the “Channel insertion loss” as given in Table 181–7.
- $MPI_and_DGD_penalties_allocation$ is the allocation for MPI and DGD penalties as given in Table 181–7.

where in 180.9.8.2

- The transmitter functional symbol error histogram mask per each lane is given in Table 180–17 and is measured using the test pattern as given in Table 181–12.

181.9.9 Transmitter overshoot and undershoot

The transmitter overshoot and undershoot of each lane shall be within the limit given in Table 181–5 if measured using the test pattern specified for transmitter overshoot and undershoot in Table 181–12.

Transmitter overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 181.9.6) and the waveform captured for the TECQ test (see 181.9.7), at the output of the reference receiver defined in 181.9.2.

Transmitter overshoot and undershoot are calculated using the methods in 140.7.7, with the exceptions that P_0 , P_3 and OMA_{outer} are measured using the methods in 181.9.5 and the hit ratio used to determine P_{max} and P_{min} is 1×10^{-4} .

181.9.10 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 181–5 if measured using the test pattern specified for transmitter power excursion in Table 181–12.

Transmitter power excursion is measured using the waveforms captured at the output of the reference receiver defined in 181.9.2.

Transmitter power excursion is defined in 140.7.8, with the exception that the hit ratio used to determine P_{max} and P_{min} is 1×10^{-4} .

181.9.11 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 181–5 if measured using one of the test patterns specified for extinction ratio in Table 181–12. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power levels P_3 and P_0 , as defined for OMA_{outer} (see 181.9.5).

181.9.12 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 181–5 if measured using the test pattern specified for transmitter transition time in Table 181–12.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. The transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 181.9.2.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 181.9.5). When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

181.9.13 Relative intensity noise (RIN_xOMA)

RIN_xOMA of each lane, with “x” referring to 17.1, which is the value for optical return loss tolerance in Table 181–5, shall be within the limit given in Table 181–5 when measured using one of the test patterns and sampling range specified for OMA_{outer} measurement in 181.9.5, but with applied x dB optical return loss and the reference receiver defined in 181.9.2. RIN_xOMA is measured using the methods specified in 180.9.13.

181.9.14 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 181–6 if measured using one of the test patterns specified for receiver sensitivity in Table 181–12.

The conformance test signal applied at TP3 meets the requirements for a 800GBASE-FR4-500 transmitter followed by an attenuator. An optical demultiplexer may be used to separate the lane having the wavelength for the lane under test as specified in Table 181–6 for calibrating the TECQ.

The TECQ of the conformance test signal is measured according to 181.9.7, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 181–6.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

The measured receiver sensitivity is the lowest value of OMA_{outer} where the PMD receiver meets the BLER requirements in 181.2, measured at the PMA using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 180–19 and $p = 4$. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20 in the column for $p = 4$. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

The Low-stressed Receiver Sensitivity (LRS) conformance test signal at TP3 meets the transmit characteristics given in Table 181–5 except for characteristics in Table 180–18.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

181.9.15 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limit given in Table 181–6 if measured using the method defined in 122.8.10, with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 181.9.6, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 181–5.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the RIN_x OMA of the SRS test source (where x is the value for optical return loss tolerance from Table 181–5) shall be no greater than the value specified in Table 181–5.
- An example stressed receiver conformance test setup is shown in Figure 122–8; however, alternative test setups that generate equivalent stress conditions may be used.
- The frequency of the sinusoidal interferer is greater than 200 MHz.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 181–5.
- The values of transmitter overshoot and undershoot and transmitter power excursion of the stressed receiver conformance test signal are within the limits specified in Table 181–5.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ OMA_{outer} of each aggressor lane” are as given in Table 181–6.

The measured stressed receiver sensitivity is the lowest value of OMA_{outer} where the receiver meets the requirements in 181.2 using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 180–19 and $p = 4$. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20 in the column for $p = 4$. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

181.10 Safety, installation, environment, and labeling

181.10.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

181.10.2 Laser safety

Optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹¹

181.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

181.10.4 Environment

Normative specifications in this clause are met by a system integrating a 800GBASE-FR4-500 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

181.10.5 Electromagnetic emission

A system integrating a 800GBASE-FR4-500 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

181.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

181.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-FR4-500).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 181.10.2.

¹¹A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

181.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 181–14 and Table 181–15. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 181–14 and Table 181–15.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 181–14—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	181.5.6	1.0.15	45.2.1.1
Global_PMD_transmit_disable	181.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_3 to PMD_transmit_disable_0	181.5.8	1.9.4 to 1.9.1	45.2.1.8

Table 181–15—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	181.5.9	1.1.7	45.2.1.2
PMD_transmit_fault	181.5.10	1.8.11	45.2.1.7
PMD_receive_fault	181.5.11	1.8.10	45.2.1.7
Global_PMD_signal_detect	181.5.4	1.10.0	45.2.1.9
PMD_signal_detect_3 to PMD_signal_detect_0	181.5.5	1.10.4 to 1.10.1	45.2.1.9

181.12 Protocol implementation conformance statement (PICS) proforma for Clause 181, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500¹²

181.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 181, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

181.12.2 Identification

181.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

181.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-20xx, Clause 181, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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¹²*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

181.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
FR4-500	800GBASE-FR4-500	181.7	Device supports requirements for 800GBASE-FR4-500 PHY	M	Yes []
*INS	Installation / cable	181.8	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
DC	Delay constraints	181.4.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	181.4.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	181.11	Registers and interface supported	O	Yes [] No []

181.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500

181.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-R PCS and PMA	181.1		M	Yes []
F2	Block error ratio	181.2	Meets the BLER specified in 181.2	M	Yes []
F3	Transmit function	181.5.2	Conveys symbols from PMD service interface to MDI	M	Yes []
F4	Optical multiplexing and delivery to the MDI	181.5.2	Optically multiplexes the four optical signals for delivery to the MDI	M	Yes []
F5	Mapping between optical signal and logical signal for transmitter	181.5.2	Optical power levels from lowest to highest correspond to tx_symbols zero, one, two, and three, respectively	M	Yes []
F6	Receive function	181.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F7	Conversion of four optical signals to symbol streams	181.5.3	For delivery to the PMD service interface	M	Yes []
F8	Mapping between optical signal and logical signal for receiver	181.5.3	Optical power levels from lowest to highest correspond to rx_symbols zero, one, two, and three, respectively	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F9	Global Signal Detect behavior	181.5.4	Global_PMD_signal_detect is an indicator of the presence of an optical signal on all lanes	M	Yes []
F10	Lane-by-lane Signal Detect function	181.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 181–2	M	Yes []
F11	PMD reset function	181.5.6	Resets the PMD sublayer	O	Yes [] No []
F12	Global transmit disable function	181.5.7	Disables all of the optical transmitters with the Global_PMD_transmit_disable variable	O	Yes [] No []
F13	PMD lane-by-lane transmit disable using management variable	181.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	O	Yes [] No []
F14	PMD lane-by-lane transmit disable using state variable	181.5.8	Disables the optical transmitter when tx_disable variable is true	M	Yes []
F15	PMD_fault function	181.5.9	Sets PMD_fault to one if any local fault is detected	O	Yes [] No []
F16	PMD_transmit_fault function	181.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	O	Yes [] No []
F17	PMD_receive_fault function	181.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	O	Yes [] No []

181.12.4.2 PMD to MDI optical specifications for 800GBASE-FR4-500

Item	Feature	Subclause	Value/Comment	Status	Support
FRF1	Transmitter meets specifications in Table 181–5	181.7.1	Per definitions in 181.9	M	Yes []
FRF2	Receiver meets specifications in Table 181–6	181.7.2	Per definitions in 181.9	M	Yes []

181.12.4.3 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	181.9	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	181.9.3	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	181.9.4	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	181.9.5	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	181.9.6	Each lane	M	Yes []
OM6	Transmitter eye closure for PAM4 (TECQ)	181.9.7	Each lane	M	Yes []
OM8	Transmitter functional symbol error histogram	181.9.8	Each lane	M	Yes []
OM8	Transmitter overshoot and undershoot	181.9.9	Each lane	M	Yes []
OM9	Transmitter power excursion	181.9.10	Each lane	M	Yes []
OM10	Extinction ratio	181.9.11	Each lane	M	Yes []
OM11	Transmitter transition time	181.9.12	Each lane	M	Yes []
OM12	Relative intensity noise	181.9.13	Each lane	M	Yes []
OM13	Receiver sensitivity	181.9.14	Each lane	M	Yes []
OM14	Stressed receiver sensitivity	181.9.15	Each lane	M	Yes []

181.12.4.4 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	181.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	181.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	181.10.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

181.12.4.5 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	181.8	Meets requirements specified in Table 181–8 and Table 181–9	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	181.8.2	Meets requirements specified in Table 181–10	INS:M	Yes [] N/A []
OC3	MDI Requirements	181.8.3	Meets IEC 61753-1 and IEC 61753-021-2	INS:M	Yes [] N/A []

181.12.4.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	181.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	181.11		MD:M	Yes [] N/A []